

EE227 – Digital Logic Design

Lecture 9

Outline

- KARANUGH Map (K – Map) Method
 - With 2 variables
 - With 3 variables
 - With 4 variables

KARANUGH Map (K – Map)

- K – Map is a pictorial method used to minimize Boolean expressions without having to use Boolean algebra theorems and equation manipulations
- It is thought as a special version of a truth table
- Using a K-map, expressions with two to four variables are easily minimized

Generating K – Map (1/2)

- A k map will contain 2^n squares
 - Where n is the number of variables in the expression
- While generating a k-map one should keep in mind that **any two adjacent squares in the map differ by only one bit**

		$x \backslash y$		0	1
x	0	m_0 $x'y'$	m_1 $x'y$		
	1	m_2 xy'	m_3 xy		

With two variables

m_0	m_1	m_3	m_2
m_4	m_5	m_7	m_6

With three variables

		$x \backslash yz$			
x	0	m_0 $x'y'z'$	m_1 $x'y'z$	m_3 $x'yz$	m_2 $x'yz'$
	1	m_4 $xy'z'$	m_5 $xy'z$	m_7 xyz	m_6 xyz'

Generating K – Map (2/2)

- With four variables

m_0	m_1	m_3	m_2
m_4	m_5	m_7	m_6
m_{12}	m_{13}	m_{15}	m_{14}
m_8	m_9	m_{11}	m_{10}

$wx \backslash yz$		00	01	11	10
		m_0 $w'x'y'z'$	m_1 $w'x'y'z$	m_3 $w'x'yz$	m_2 $w'x'yz'$
00	m_4 $w'xy'z'$	m_5 $w'xy'z$	m_7 $w'xyz$	m_6 $w'xyz'$	
01	m_{12} $wxy'z'$	m_{13} $wxy'z$	m_{15} $wxyz$	m_{14} $wxyz'$	
11	m_8 $wx'y'z'$	m_9 $wx'y'z$	m_{11} $wx'yz$	m_{10} $wx'yz'$	
10					

K-Map - Rules of Simplification (1/4)

- While using k-map we group the 1's present in the k-map table and note the variables which are not changing
- Here are some rules for simplification and grouping of 1's
 - Groups may not include any cell containing a zero
 - Groups may be horizontal or vertical, but not diagonal

A \ B	0	1
0	0	
1	1	

WRONG ✗

A \ B	0	1
0	0	
1	1	1

RIGHT ✓

A \ B	0	1
0	0	1
1	1	0

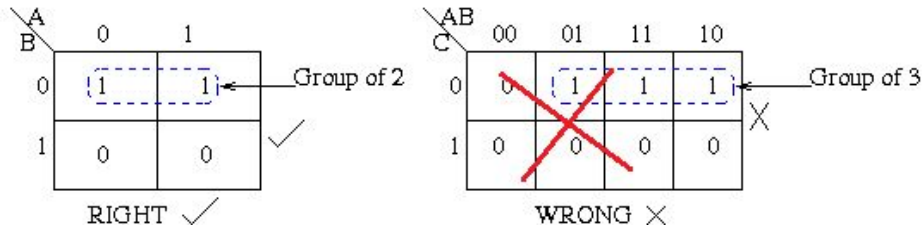
WRONG ✗

A \ B	0	1
0	0	1
1	1	1

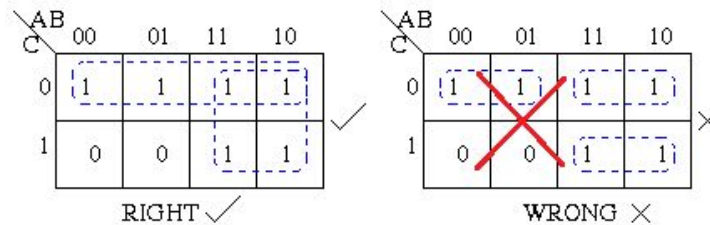
RIGHT ✓

K-Map - Rules of Simplification (2/4)

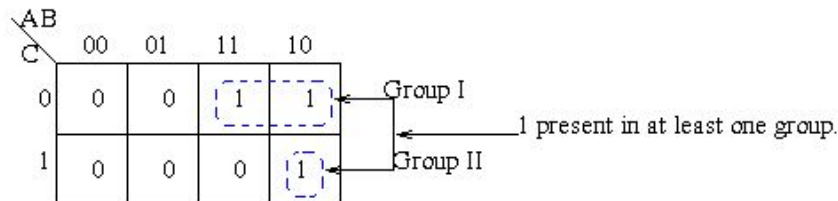
- Groups must contain 1, 2, 4, 8, or in general 2^n cells



- Each group should be as large as possible

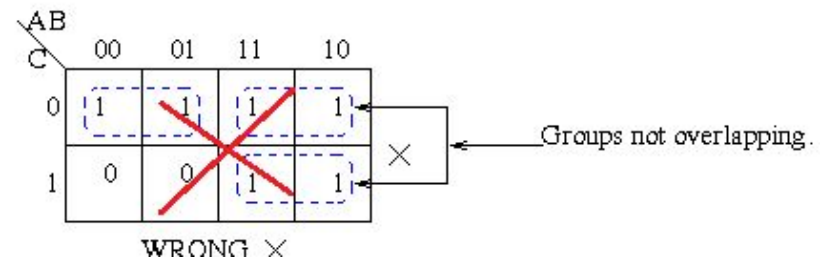
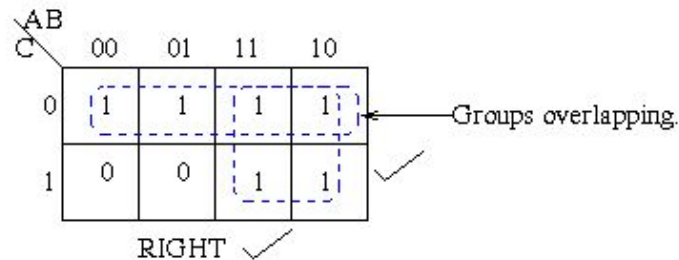


- Each cell containing a one must be in at least one group

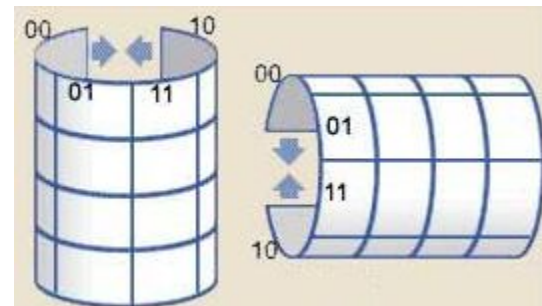
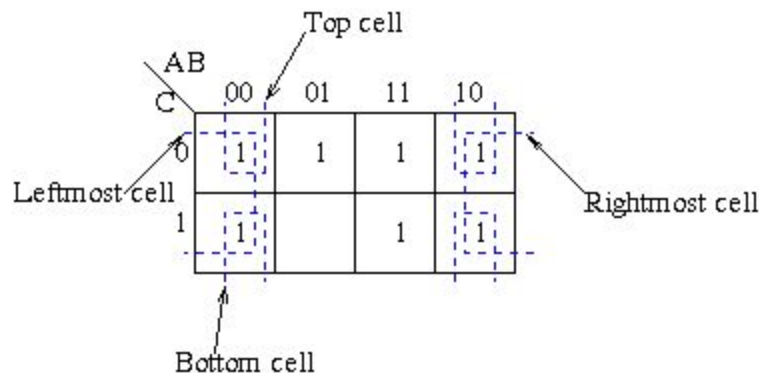


K-Map - Rules of Simplification (3/4)

- Groups may overlap

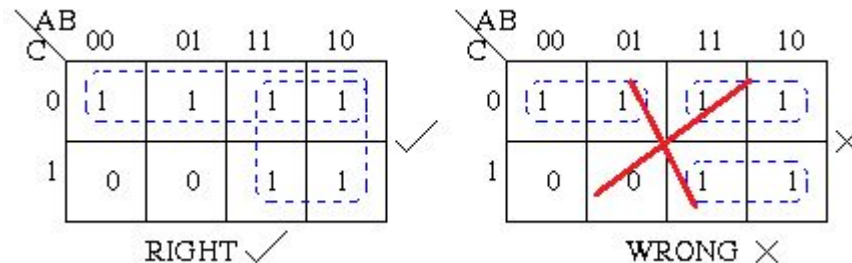


- Groups may wrap around the table. The leftmost cell in a row may be grouped with the rightmost cell and the top cell in a column may be grouped with the bottom cell



K-Map - Rules of Simplification (4/4)

- There should be as few groups as possible, as long as this does not contradict any of the previous rules



K – Map Simplification Steps

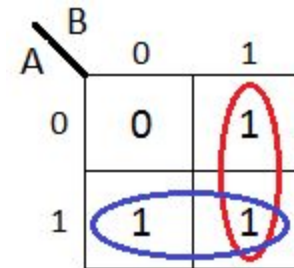
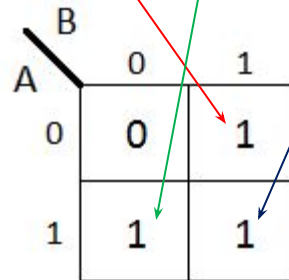
1. Populate the k-map
 - By using truth table
 - Or by using minterms
 - Introduce all the variables in each term
2. Form the groups by considering the rules
3. Note the constant variables for each group
4. Multiply the constant variables of each group to form a term
5. Add all the terms to obtain the simplified Boolean function

K – Map with Two Variables

Simplify the following expression by using k-map

$$F = A'B + AB' + AB$$

A	B	A'	B'	A'B	AB'	AB	F
0	0	1	1	0	0	0	0
0	1	1	0	1	0	0	1
1	0	0	1	0	1	0	1
1	1	0	0	0	0	1	1



The simplified expression is

$$F = A + B$$

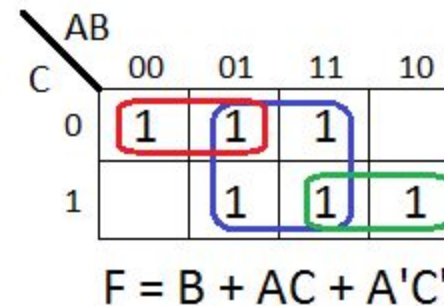
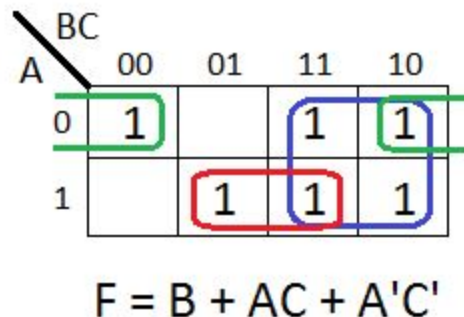
K – Map with Three Variables

Simplify the following Boolean function by using k-map

$$F = A'B'C' + A'B + ABC' + AC$$

$$F = A'B'C' + A'B(C + C') + ABC' + AC(B + B')$$

$$F = A'B'C' + A'BC + A'BC' + ABC' + ABC + AB'C$$

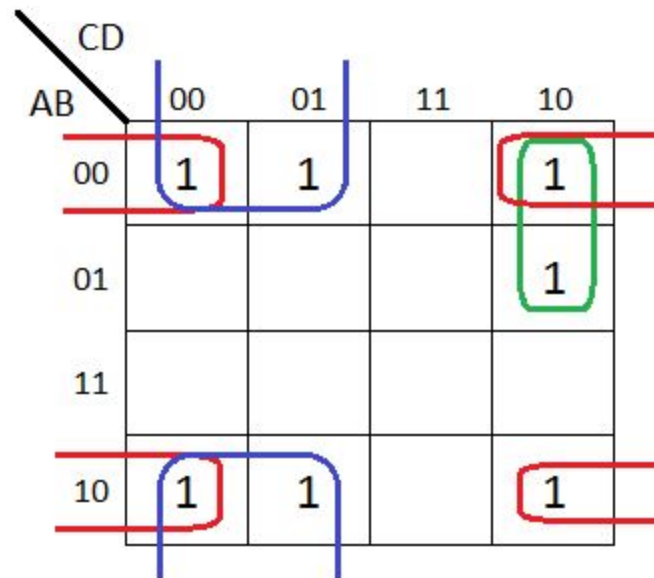


K – Map with Four Variables (1/2)

Simplify the following Boolean function by using k-map

$$F = A'B'C' + B'CD + A'BCD' + AB'C'$$

$$F = A'B'C'D + A'B'C'D' + AB'CD + A'B'CD + A'BCD' + AB'C'D + AB'C'D'$$



$$F = B'D' + B'C' + A'CD'$$

K – Map with Four Variables (2/2)

Simplify the following Boolean function by using k-map

$$F = A'B'C'D' + AC'D' + B'CD' + A'BCD + BC'D$$

$$F = A'B'C'D + ABC'D' + AB'C'D' + A'B'CD' + AB'CD' + A'BCD + A'BC'D + ABC'D$$

		CD			
		00	01	11	10
AB	00	1			1
	01		1	1	
	11	1	1		
	10	1			1

$$F = B'D' + ABC' + A'BD$$

Practice Problem 1

- Simplify the following Boolean functions by using k-map method
 - a) $F_1 = xy + x'y'z' + x'yz'$
 - b) $F_2 = A'B + BC' + BC + AB'C'$
 - c) $F_3 = M + MC + AC + AM + AMC$
 - d) $F(W,X,Y,Z) = \sum (1, 4, 5, 6, 12, 14, 15)$

Answers

- a) $F1 = x'z' + xy$
- b) $F2 = B + AC'$
- c) $F3 = M + AC$
- d) $F(w,x,y,z) = w'y'z + wxy + xz'$